From Algorithm to the Clinic: Designing a Usable Clinical Decision Support (CDS) Interface and EHR Integration Strategy

Team HAC

Roadmap (1/1)

- Research Question – *Why are we doing this?*
- Model Pipeline - *How We Find Answers?*
- Data exploratory and Preprocessing- *EDA drives design*
- Key method – *Why we can trust the pipeline*
- Causal Models - *How We Compare Models*
- Clinical Heterogeneity – *One treatment does NOT fit all patients?*
- Conclusions

We would like to thank the authors of the papers referenced in this study.

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- **Austin, P. C.** (2021). *Missing data in clinical research: A tutorial on multiple imputation*. U.S. National Library of Medicine. https://pmc.ncbi.nlm.nih.gov/articles/PMC8499698/
- **Djupskås, A., Stasik, A. J., & Riemer-Sørensen, S**. (2025). *Unreliable uncertainty estimates with Monte Carlo dropout* (arXiv preprint arXiv:2512.14851). https://arxiv.org/abs/2512.14851
- **Lee, K. M., & Emsley, R. (2024).** The impact of heterogeneity on the analysis of platform trials with normally distributed outcomes. *BMC Medical Research Methodology, 24*, Article 163. https://doi.org/10.1186/s12874-024-02293-4
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- **Yang, J., Wan, Y., Ni, Q., et al. (2022).** Quantifying causal effects from observed data using quasi-intervention. *BMC Medical Informatics and Decision Making, 22*, 337. https://doi.org/10.1186/s12911-022-02086-z
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- **Zhao, Z., Zhang, Y., Harinen, T., & Yung, M. (2022).** Feature selection methods for uplift modeling and heterogeneous treatment effect. In *Proceedings of the 18th IFIP International Conference on Artificial Intelligence Applications and Innovations (AIAI)* (pp. 217–230). Springer. https://doi.org/10.1007/978-3-031-08337-2_19

How can patient-specific treatment effects of surgery versus conservative care be estimated to support shared decision making?

For outcome $k \in \{1, \dots, 5\}$*:*

$$TE_k(x) = \mathbb{E}[Y_k(1) - Y_k(0) \mid X = x]$$

Model pipeline (1/1)

Data Preparation and Feature Engineering

- Explore Data
- Missing handling
- BMI Computation
- Feature Selection
- Data Pre-processing

Propensity Score and Covariate Balance

- Propensity Estimation
- Covariate Balance Diagnostics

Causal Meta-Learner Implementations

- S-Learner
- T-Learner
- X-Learner
- R-Learner
- MLP within a T-learner framework.

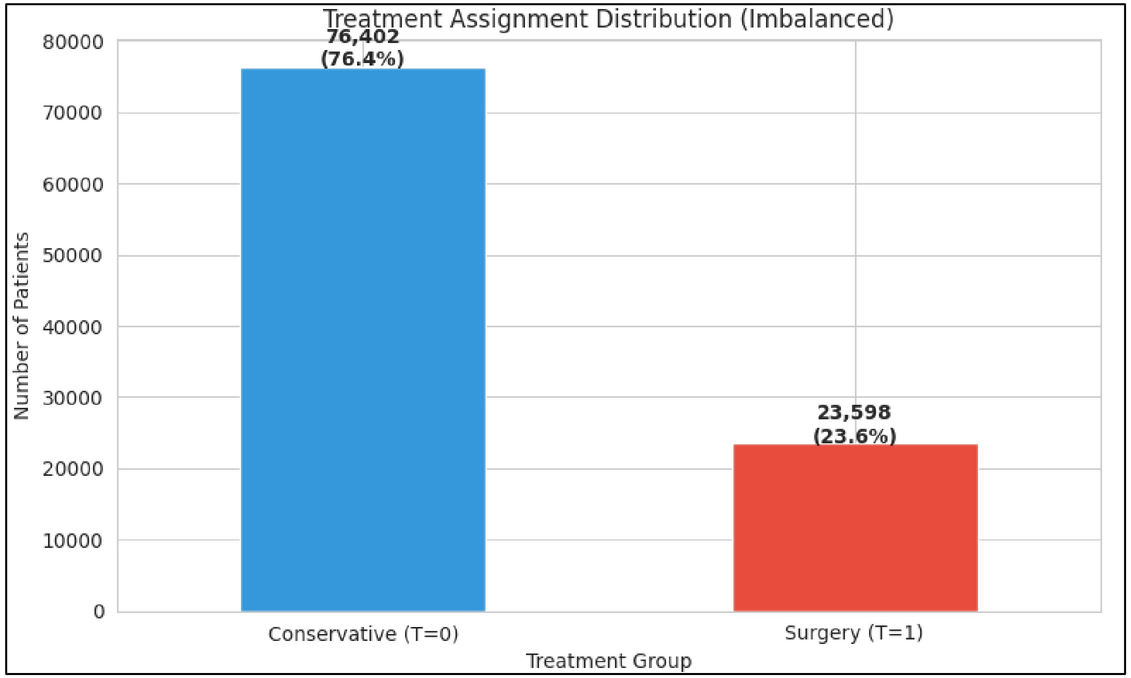
Model Selection and Evaluation

- 5-fold cross-validation
- Inverse Transformation TE5
- R-loss, MAE, RMSE

Clinical Heterogeneity and Final Outputs

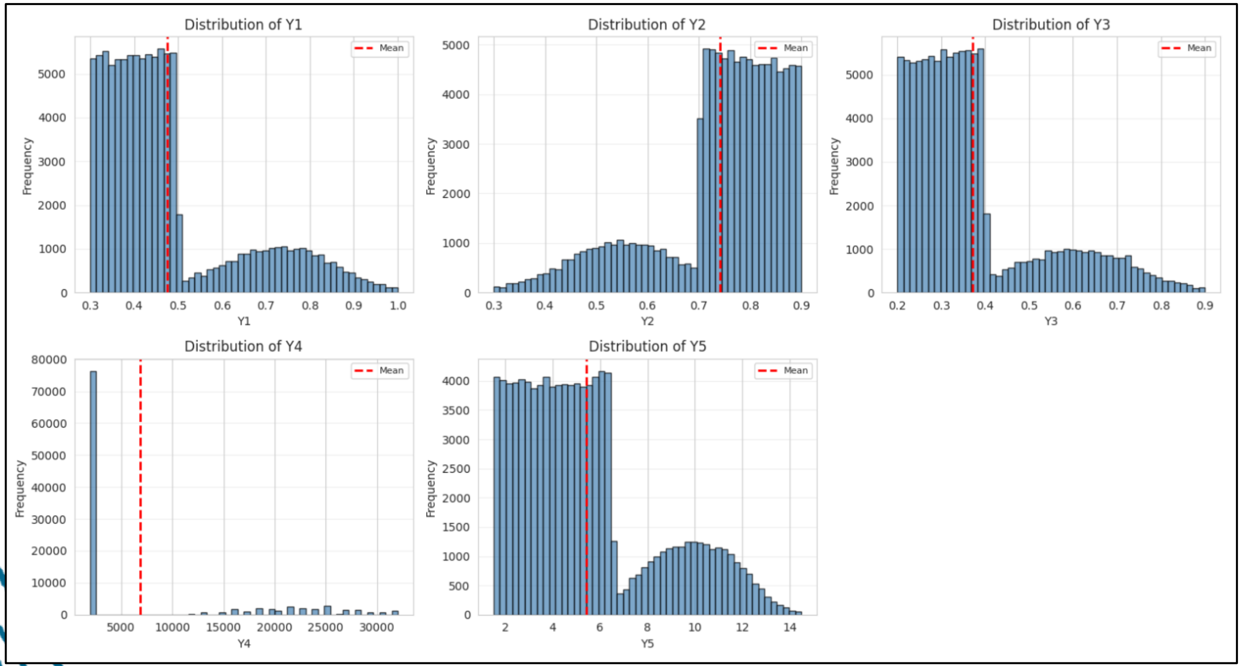
- Driver Analysis
- CDS-Ready Table

Data exploratory and Preprocessing (1/4)



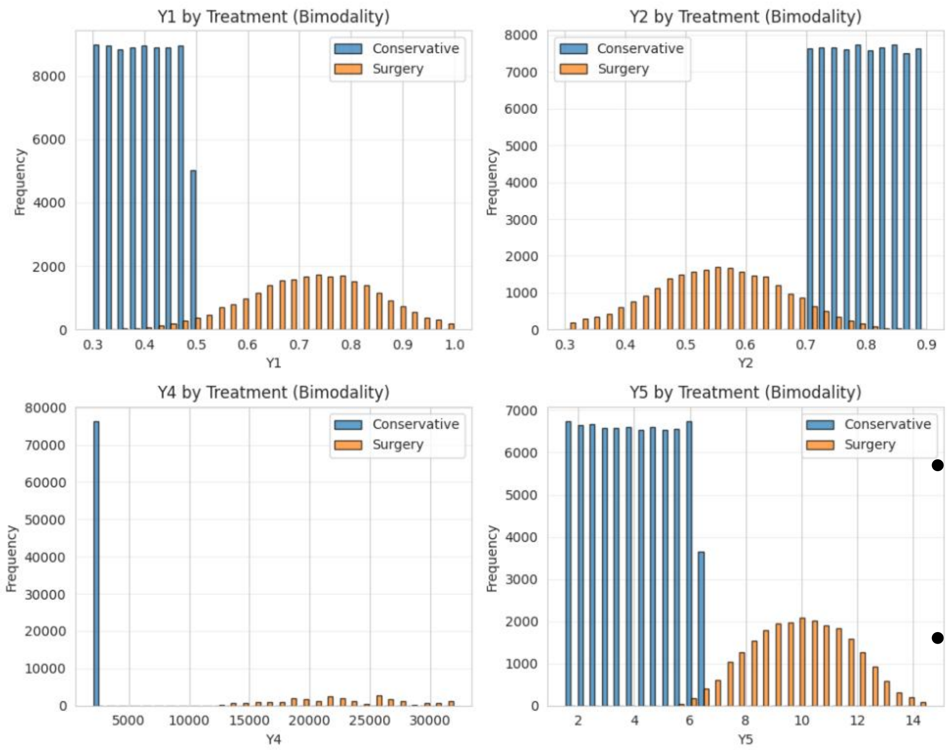
- **EDA Insight:** Treatment assignment is highly imbalanced (24% surgery).
- **Implication:** Strong selection bias; naive outcome comparisons are invalid.
- **Solution:** Propensity score modeling with IPW and balance diagnostics.

Data exploratory and Preprocessing (2/4)



- **EDA Insight:** Outcomes exhibit strong skewness and bimodality.
- **Implication:** Treatment effects are heterogeneous and non-linear.
- **Solution:**
  - Log1p transformation for cost and time outcomes.
  - Causal meta-learners with outcome-specific modeling.
  - R-loss.

Data exploratory and Preprocessing (3/4)



- **EDA Insight:** Outcome distributions differ substantially by treatment group.
- **Implication:** Differences may reflect causal effects or confounding.
- **Solution:** Doubly robust causal learning with balance diagnostics.

Data exploratory and Preprocessing (4/4)

50 columns with missing values:

Column	Missing_Count	Missing_Pct	Dtype
X51	72717	72.72	float64
X54	3081	3.08	object
X60	3081	3.08	object
X21	3069	3.07	object
X20	3055	3.06	float64
X15	3055	3.06	float64
X16	3056	3.06	object
X41	3053	3.05	object
X14	3050	3.05	datetime64[ns]
X40	3047	3.05	float64
X26	3046	3.05	float64
X36	3050	3.05	float64
X38	3043	3.04	object
X13	3042	3.04	float64
X59	3038	3.04	object
X18	3041	3.04	datetime64[ns]
X57	3043	3.04	float64
X45	3028	3.03	object
X24	3031	3.03	float64
X30	3031	3.03	float64
X35	3029	3.03	float64
X48	3019	3.02	object
X62	3022	3.02	float64
X37	3015	3.02	float64
X44	3015	3.02	object
X50	3014	3.01	object
X58	3004	3.00	float64

- **EDA Insight:** No Duplicate, Substantial and variable missingness across baseline features.
- **Implication:** Dropping data would introduce bias and reduce power.
- **Solution:** Structured imputation with explicit missingness indicators.

Key Methodology (1/5)

Uplift-aligned F-filter

Why:
Help model focusing on who benefits more rather than just predicting outcomes.

Definition
Uplift-aligned F-filter is a feature-screening method that keeps only variables that change the treatment–control difference (individualized treatment effect).

Core formulation
For each feature X_j ,compute its interaction strength with treatment:

$$F_j = \frac{Var(\mathbb{E}[Y' \mid X_j T = 1] - \mathbb{E}[Y' \mid X_j T = 0])}{Residual\ variance}$$

Interpretation:

- High F_j: feature changes the gap between treatment and control → keep it
- Low F:$_j$ feature predicts outcome but not treatment effect → drop it

Key Methodology (2/5)

Propensity Score Weighting by Logistic Regression Model + Balance Diagnostics

Why:
Balance diagnostics justify causal estimation by confirming that treatment groups are comparable.

Definition
Propensity score weighting (IPW) reweights observational data so that treated, and control groups become comparable with respect to baseline clinical covariates, approximating a randomized experiment.

Core formulation
Propensity score:

$$e(X) = P(T = 1 \mid X)$$

Stabilized inverse probability weights:

$$w_i = \begin{cases} \dfrac{P(T=1)}{e(X_i)} & \mathit{if}\ T_i = 1 \\ \dfrac{P(T=0)}{1 - e(X_i)} & \mathit{if}\ T_i = 0 \end{cases}$$

Balance diagnostic (Standardized Mean Difference):

$$\mathrm{SMD} = \frac{\bar{X}_{T=1} - \bar{X}_{T=0}}{s_{pooled}}$$

Key Methodology (3/5)

Causal Meta-Learners (S, T, X) with Tree-based models

Why:
Meta-learners translate heterogeneous clinical responses into patient-specific causal effects.

Definition
Meta-learners are flexible strategies that estimate individualized treatment effects by modeling potential outcomes under each treatment condition.

Core formulation
Potential outcome models:

$$\mu_1(x) = \mathbb{E}[Y' \mid T = 1X = x], \mu_0(x) = \mathbb{E}[Y' \mid T = 0X = x]$$

Individual treatment effect:

$$\tau(x) = \mu_1(x) - \mu_0(x)$$

Key Methodology (4/5)

Doubly Robust / R-Learner

Why:
Doubly robust learning protects causal estimates against residual confounding and model error.

Definition
The **R-learner** isolates treatment effect heterogeneity by residualizing both the outcome and the treatment assignment, making it robust to model misspecification.

Core objective

$$\min_{\tau(\cdot)} \quad \mathbb{E}[(Y - \hat{m}(X) - (T - \hat{e}(X))\tau(X))^2]$$

Where:
$\hat{m}(X)$is the outcome regression
$\hat{e}(X)$is the propensity score

Key Methodology (5/5)

Cross-Fitting + R-Loss Model Selection

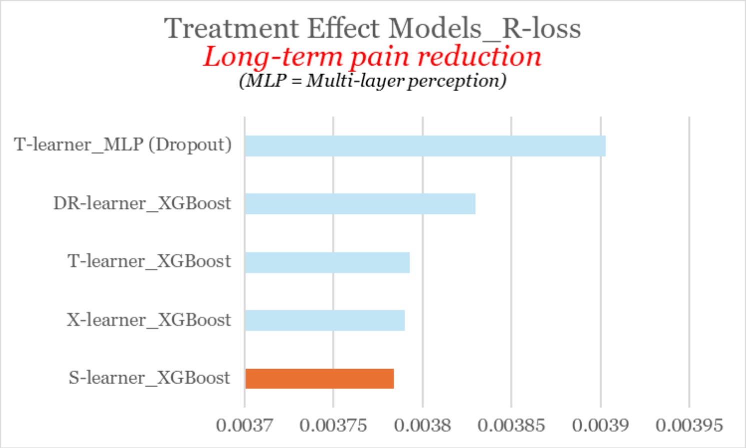
Why:
Cross-fitted R-loss ensures that the selected model best estimates causal effects, not just predictions.

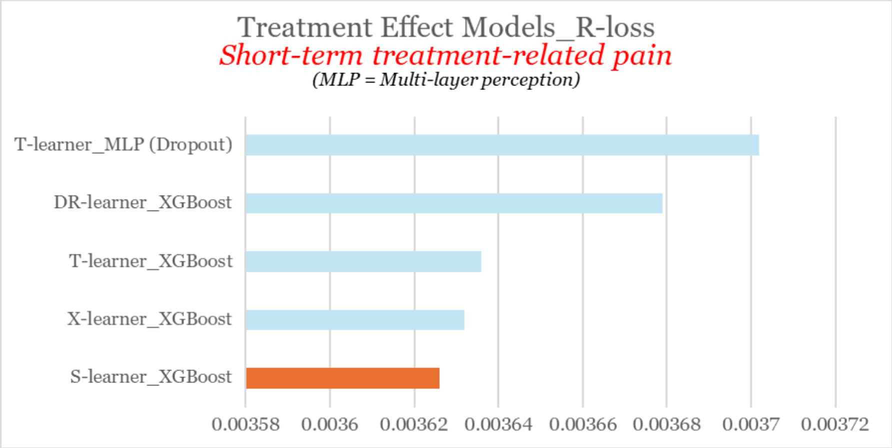
Definition
Cross-fitting separates nuisance estimation from treatment effect estimation, while **R-loss** evaluates treatment effect quality without observing counterfactuals.

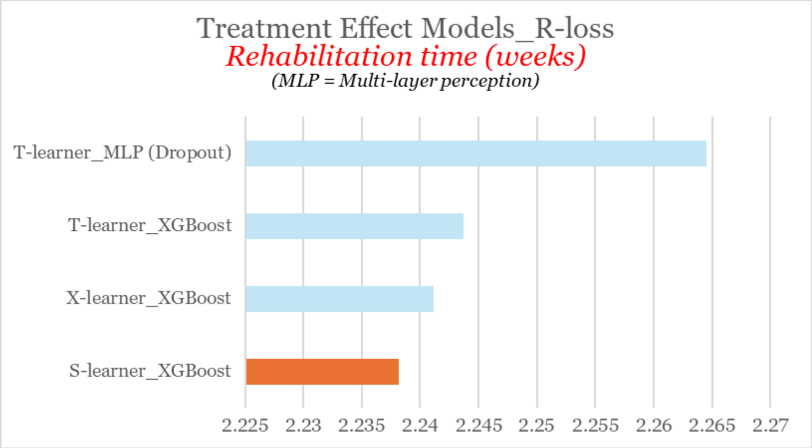
Core formulation (R-loss)

$$\mathcal{L}_R = \mathbb{E}[(Y - \widehat{m}(X) - \big(T - \hat{e}(X)\big)\hat{\tau}(X))^2]$$

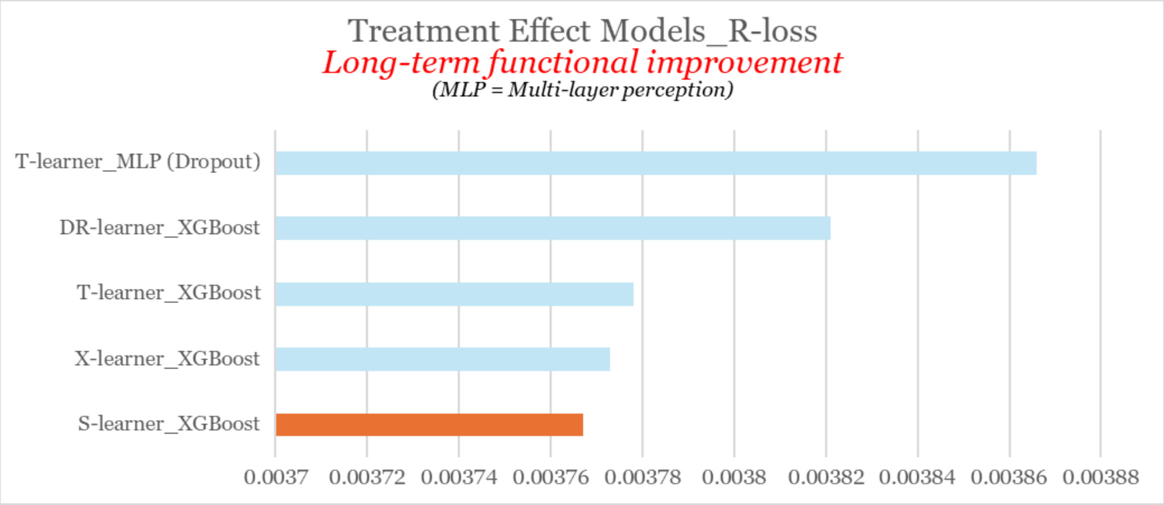
Causal Models *(1/2)*

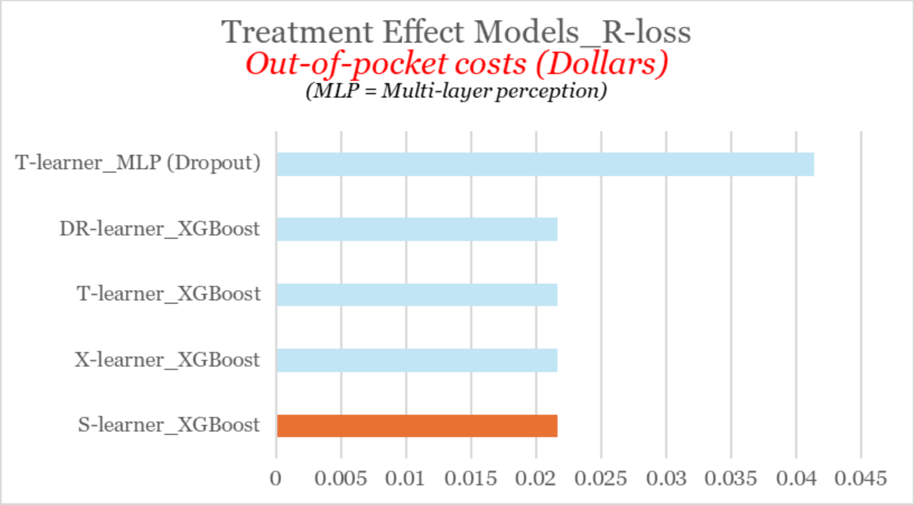






S-learner gives the most accurate individualized treatment effects





Causal Models *(2/2)*

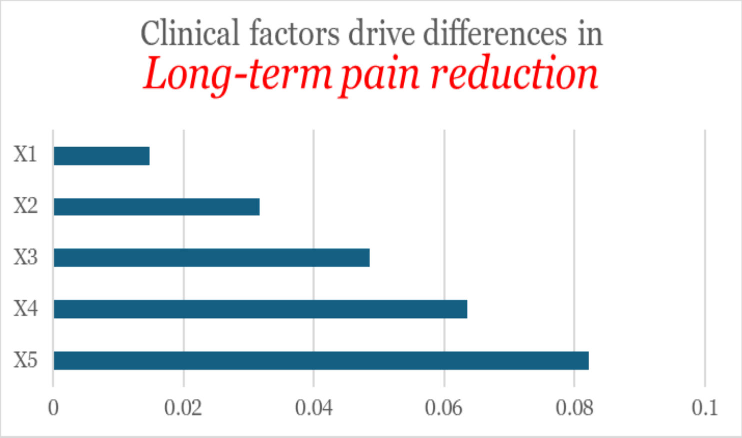
Outcome	Arm	MAE	RMSE
Y1	T=1	0.050	0.058
Y1	T=0	0.050	0.058
Y2	T=1	0.050	0.058
Y2	T=0	0.050	0.058
Y3	T=1	0.050	0.058
Y3	T=0	0.050	0.058
Y4	T=1	25.325	29.446
Y4	T=0	25.031	28.916
Y5	T=1	1.254	1.449
Y5	T=0	1.257	1.451

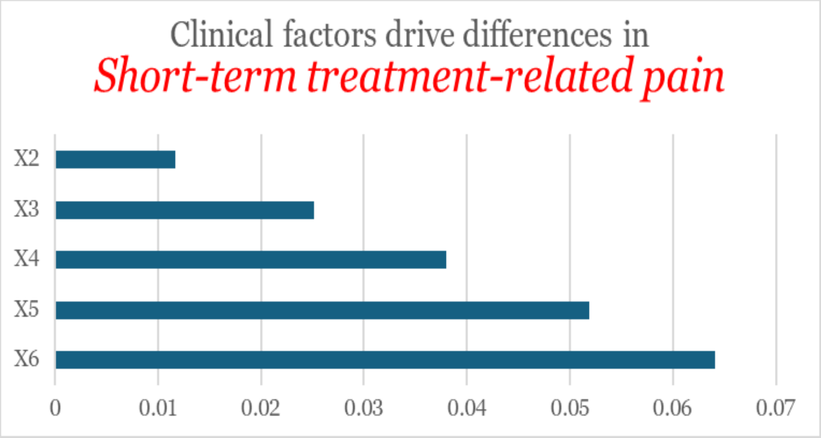
- **Model accuracy is balanced across treatment arms**
- **Predictions are clinically reasonable**
  Pain & function: ~0.05
  Rehab time: ~1 week
  Cost: ~$25

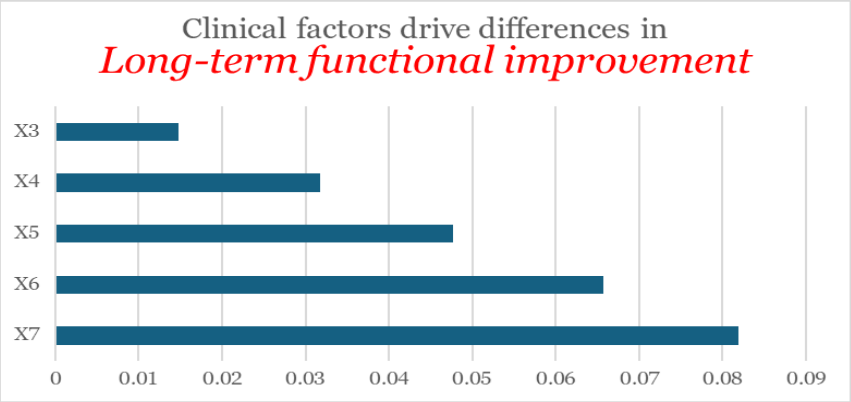
Clinical Heterogeneity

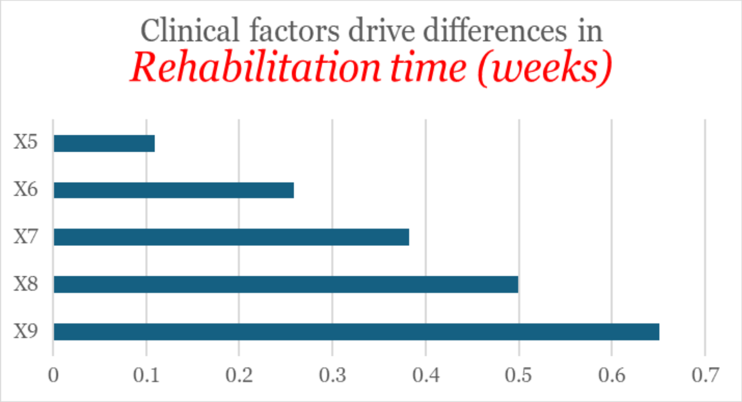
- The same surgery leads to very different benefit–cost profiles across patients.
- Surgery does not benefit everyone in the same way
- Different baseline clinical factors drive pain relief, functional recovery, rehabilitation time, and cost
- The need for individualized, preference-aware treatment decisions.

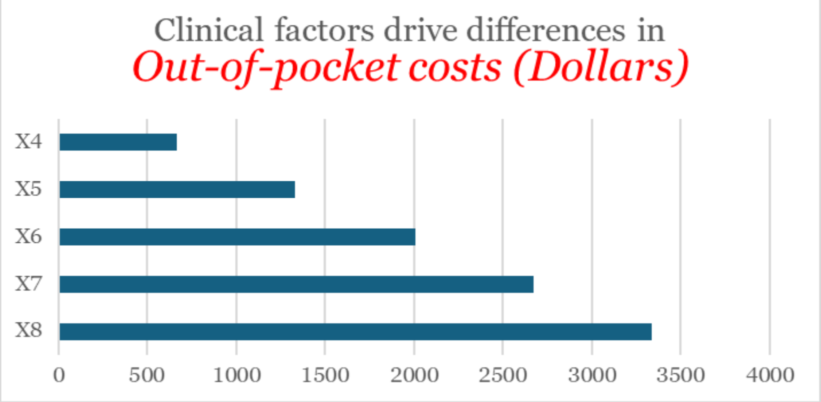
Outcome	TE Range
Pain Reduction	−0.01 to +0.53
Short-term Pain	−0.42 to -0.00
Functional Improvement	−0.01 to +0.53
Cost Impact (USD)	+$10k to +$30k
Rehab Time (weeks)	+3.2 to +8.7











Conclusion

*The **S-learner with XGBoost Model** can be used to compare treatment options and select the one that best fits a patient's condition and priorities.*

Thank you